Chapter 4
Introduction to Statistics and Probability



This chapter provides an overview of statistics and probability concepts that are used in metrology. It begins with a discussion on data types along with Exploratory Data Analysis techniques that can be used to help understand and visualize data. It then gives an overview of probability distributions common to metrology with guidance on choosing probability distributions based on available information. This chapter concludes with information on estimating parameters and assessing goodness-of-fit of probability models.

4.1 Introduction

The practice of metrology relies heavily on a variety of statistical methods used to design, characterize, and quantify the uncertainty of measurement systems. In this section we provide an overview of basic probability and statistics concepts that are often used to this end.

In statistics for metrology, the primary goal is to draw conclusions (e.g., estimate parameters) about a target population (measurement system) using a finite sample from that population. In metrology the parameters of concern are usually uncertainty terms, expressed as standard deviations. Once uncertainty terms have been estimated, probability statements can be made regarding future measurements. The probability statement will usually be in the form of an interval that is expected to contain the true value of a measurand.

This chapter will first introduce different types of data and will then discuss statistical methods that can be used to describe sample data. It will end with an overview of common probability distributions used in metrology, as well as guidance for estimating parameters of those distributions and assessing the goodness-of-fit of probability models.

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S. Crowder et al., *Introduction to Statistics in Metrology*,
https://doi.org/10.1007/978-3-030-53329-8_4

4.2 Types of Data

A first step before analysis is to identify the type of data that was collected and the methods that may be appropriate for that type of data. Data can be divided into two broad categories: continuous and discrete. Continuous variables, such as temperature and voltage, can take on any value in a continuous interval. Alternatively, discrete variables can only take on distinct, separate values. There are several types of discrete data, including:

- Binary data, which can fall into one of two categories. Commonly, binary data are described as "Pass/Fail", "0/1," or "Go/No Go." In metrology, it may be of interest to evaluate the performance of a "Go/No Go" gauge used in the inspection of manufactured parts.
- Count data, in which the data are nonnegative integers. The integer values come from a counting process rather than from measuring some characteristic on a continuous scale. An example is using a neutron detector to count the number of neutrons produced by a manufactured neutron source.
- Categorical data, in which the data fall into one of several distinct, nonnumeric categories. Categorical data can be further divided as ordinal or nominal. Ordinal data has an intrinsic ordering (such as low, medium, and high), while nominal data is nonordered (such as Tester A vs. Tester B).

In metrology, the data are most likely to be continuous variables, although we will present examples of assessing uncertainty using both binary (Sect. 11.3) and count data (Sect. 6.4.1).

It is important to identify the variable type as it affects the exploratory data analysis (Sect. 4.3) and choice of probability distribution (Sect. 4.4). Note that continuous data provide much more information than discrete data, so it is advantageous to use continuous measures whenever possible. For example, if instead of categorizing temperature using ratings such as "low," "medium," and "high," the measurements are on a continuous scale, then more information is available to estimate population parameters such as the mean or standard deviation.

4.3 Exploratory Data Analysis

Once the data type has been determined, Exploratory Data Analysis (EDA) can be used to better understand the overall characteristics of the data. In particular, EDA can be helpful for

- Identifying potential outliers and trends present in the data. Outliers can inflate the uncertainty standard deviations. If an "assignable cause" of an outlier can be identified and removed from the measurement process, the outlier can be discounted. Trends over time that are identified in the measurement process must be accounted for when estimating uncertainty intervals that are expected to be valid over time.

- Determining an appropriate probability distribution. The validity of uncertainty intervals will depend on correctly specifying the underlying probability distribution.
- Detecting subpopulations in the data. Subpopulations (i.e., a subset of measurements taken under very different conditions than the bulk of the data) can have the effect of inflating the measurement uncertainty. Subpopulations may need to be discounted or analyzed separately.
- Generating summary statistics. Summary statistics (e.g., sample means and standard deviations) are inputs to uncertainty analyses for both direct and indirect measurements (Chap. 6).

Commonly, EDA consists of two main parts: calculating summary statistics and creating graphical displays of the sampled data.

4.3.1 Calculating Summary Statistics

A statistic is simply a measure of some attribute of sampled data. There are many summary statistics that can be used to concisely describe features of data. Generally, for continuous data, measures of the distribution's center (e.g., mean, median or mode) and spread (e.g., range, standard deviation, inter-quartile range) are used. For data sets with multiple continuous variables, the relationship between variables (measured by the correlation coefficient) is often of interest in metrology studies. The following sections describe each of these measures in more detail. A discussion of summary statistics for discrete data follow below.

4.3.1.1 Summary Statistics for Continuous Data

Measures of Central Tendency Measures of central tendency are often used to quantify where the mid-point or center of sampled data falls. The most commonly used measure is the sample mean, also known as the sample average, which is simply the sum of the data values divided by the number of observations n. Formally, for observations $x_1, x_2, \ldots, x_n$ it is defined as

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i. \tag{4.1}$$

Another common measure of central tendency is the median, defined as the 50th percentile of the data (i.e., the point below which 50% of the observations fall). The mode is the most frequently observed value in the sampled dataset and can also be used as a measure of central tendency.

where n is the sample size, and x_i, y_i are the individual measurements of variable x and y. The formula assumes a natural pairing of the x and y variables. Values of r_{xy} close to zero indicate little to no correlation, while values of $|r_{xy}|$ close to one suggest strong correlation (positive or negative) between the variables. If variables that appear in a measurement equation (Eq. (6.27)) are highly correlated, the total variance (Eq. (7.25)) associated with the measurement equation must include additional terms to account for the correlation. This component of the EDA is thus an important consideration in any complex measurement system uncertainty analysis.

4.3.1.2 Summary Statistics for Discrete Data

For discrete categorical data, common summary statistics are the frequency and relative frequency of each category in which the data fall. Frequency is defined as the number of values in a category, while relative frequency is defined as the frequency divided by the total number of data points.

The values in a frequency table provide a rough estimate of how the data are distributed. Table 4.1 provides an example of the frequencies and relative frequencies for binary data. In this example most of the data are classified as Pass. Again, discrete data inherently provide less information than continuous data, so it is preferable to use a continuous measure if possible.

For discrete count data, the summary statistics described above for continuous variables data still apply, but the interpretation of the sample means and standard deviations may be different depending on the specific distribution used to describe the data. For example, the sample mean (Eq. (4.1)) of count data assumed to be from a Poisson distribution (Sect. 4.4.1.2) estimates both the mean and variance of the distribution. Interpretation of the summary statistics in these cases must thus consider the particular discrete probability distribution.

4.3.2 Graphical Displays of Data

In addition to summary statistics, it can be useful to visualize data to assess if there are significant outliers, trends, correlations, subpopulations, or unusual variation present. Graphical tools can also be used to suggest which probability distribution may be appropriate for the measurement variables. The choice of graphical display also depends on the type of data.

Table 4.1 Frequency and relative frequency for discrete data

Category	Frequency	Relative frequency
Pass	15	15/20 = 0.75
Fail	5	5/20 = 0.25
Total	20	20/20 = 1.00

4.3.2.1 Graphical Displays for Continuous Data

Histograms Histograms are generated by first dividing the range of the variable into equally spaced intervals, and then tabulating the number of data points that fall within each interval. The height of the histogram bar is equal to the proportion of data in its corresponding interval. Histograms provide a visualization of the distribution of continuous data including its central tendency and spread. The histogram can also be used to identify outliers and subpopulations and help determine an appropriate probability distribution.

Note that histograms can be sensitive to the choice of bin width; that is, the size of the interval spanned by the bar. Smaller bin widths will produce a more-refined version of the histogram, while larger bin widths will produce a less-refined histogram. The choice of bin width can be particularly important when there are subtle characteristics in the data distribution. The left and right plots of Fig. 4.1 show a histogram with small and large bin widths, respectively. The data are centered around zero and they appear to be normally distributed. Also included in these plots is a smoothed estimate of the histogram, shown as a black line. This smoothed estimate was created using Kernel Density Estimation (KDE), which is a nonparametric approach to estimating a distribution of data. KDE can be performed in most statistical software packages; for more information, see Silverman (1986).

Boxplots Boxplots provide a simple way to visualize the central tendency and spread of a dataset. Four characteristics can be assessed from a boxplot:

- The location of the median.
- The spread of the data.
- Whether the distribution of the data is skewed or symmetric.
- Whether there are potential outliers.

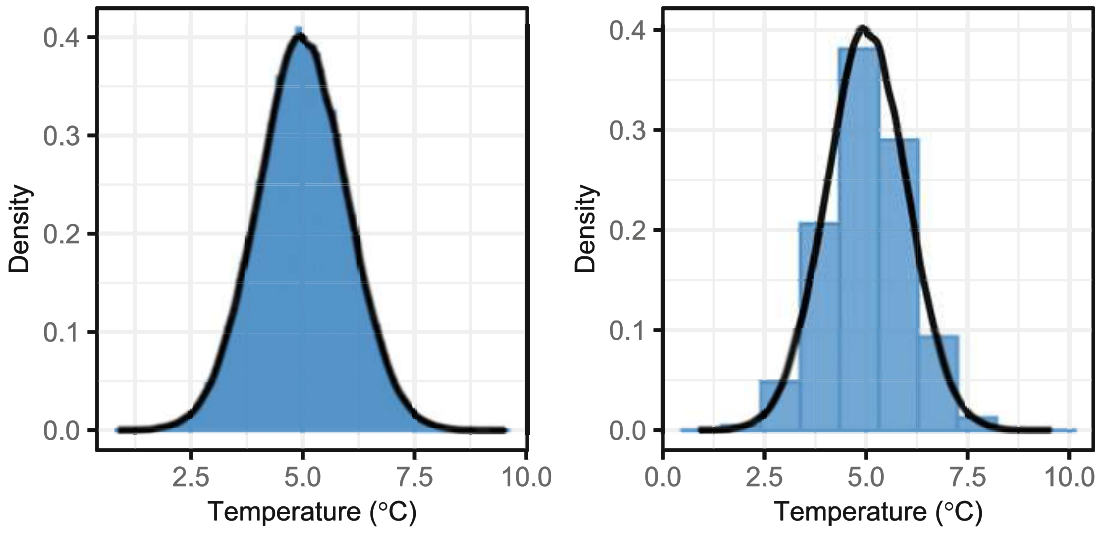


Fig. 4.1 Histograms with varying bin-widths. The left plot shows a smaller bin width (and a more-refined histogram), while the right plot shows a larger bin width (and a less-refined histogram)

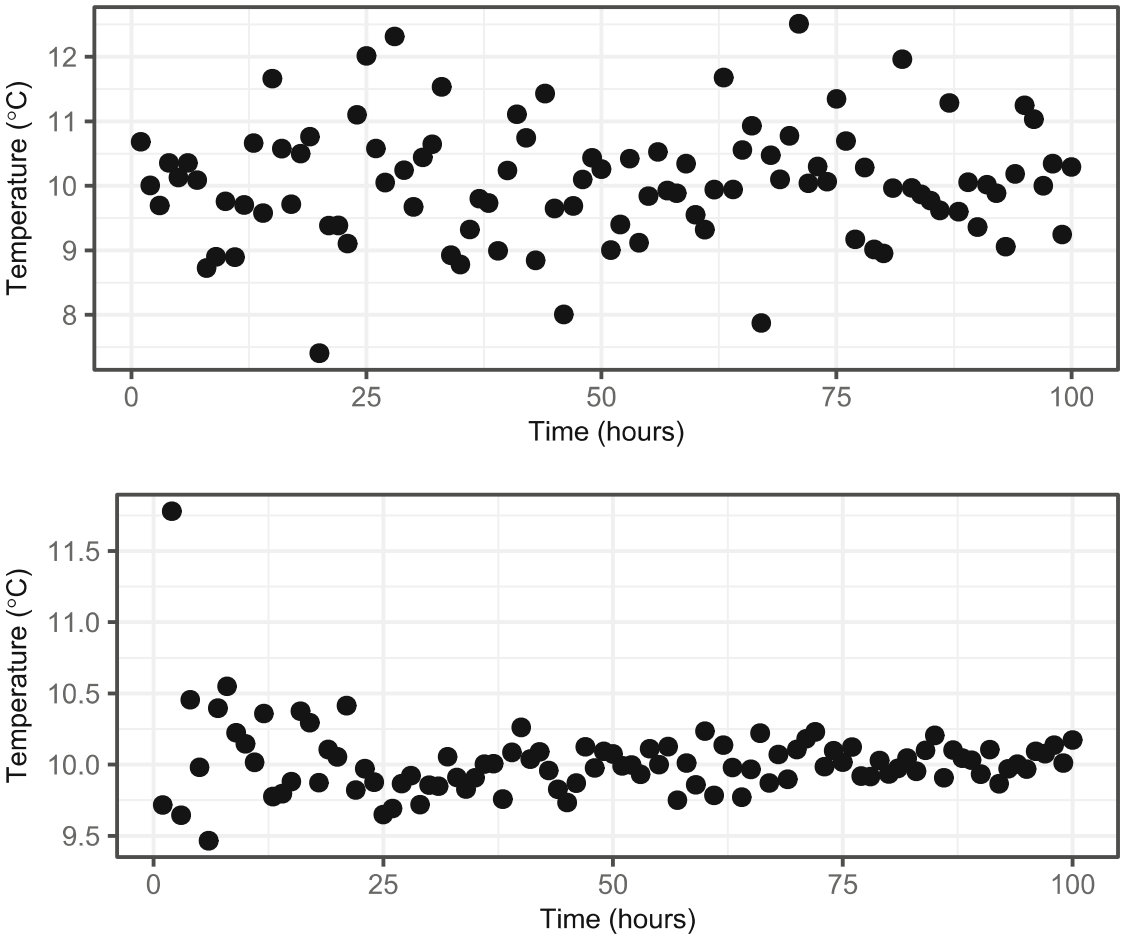


Fig. 4.3 Example of scatterplots of temperature data across time. The top plot shows no observable trend, while the bottom plot shows data with decreasing variance over time

The graphs below (each plotting a y variable versus an x variable) show examples of both low correlation (r_{xy} near zero) and high correlation ($|r_{xy}|$ near 1) (Fig. 4.4).

This component of the EDA can be used to help determine what terms should be included in the expression for the total variance (Eq. (7.25)) in an uncertainty analysis.

4.3.2.2 Graphical Displays for Discrete Data

For discrete data, the most commonly used graphical display is a frequency plot (also known as a bar plot). A frequency plot is similar to a histogram; the primary difference is that bars are given for each category or level of the discrete variable. The height of the bar is equal to the number of data points that fall into that category. Figure 4.5 gives a frequency plot with the discrete variable (temperature) on the x-axis and the count on the y-axis.

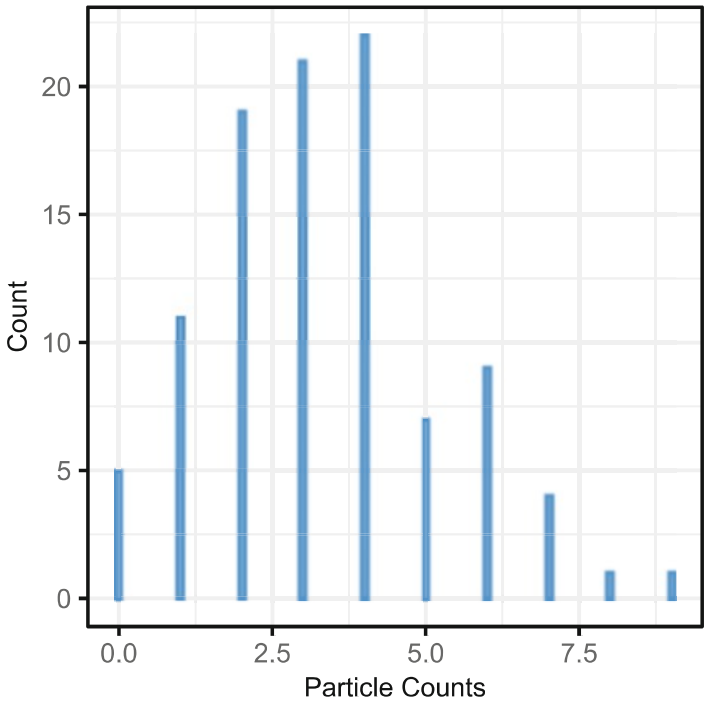


Fig. 4.6 Example frequency plot for discrete probability distribution

data were sampled. This distribution predicts future events based on the parameters estimated for the population.

4.4 Probability Distributions

Probability distributions are used in metrology to express the uncertainty in both direct measurements and in the input variables of an indirect measurement equation (Chap. 6). These input variables can be considered random variables, defined in ISO 3534-1 1993 as "a variable that may take any of the values of a specified set of values and with which is associated a probability distribution." A probability distribution is the function that assigns probabilities to the various outcomes of a random variable. Probability distributions are defined differently for discrete and continuous random variables. For a continuous random variable, a Probability Density Function (PDF) is used to assign probabilities to intervals of potential outcomes, while for a discrete random variable, a Probability Mass Function (PMF) assigns probabilities to potential outcomes that are exact values (Casella and Berger 2002).

The characteristics of a probability distribution are determined by its parameters. Most distributions have between one and three parameters. For example, the location and spread of a normal distribution are determined by its two parameters: μ, the mean, and σ^2, the variance. For a normal distribution, these parameters can be estimated directly from the data using the sample mean and the sample variance (as defined above). For other distributions, parameters can be estimated using a variety of techniques, as discussed in Sect. 4.4.2.

The three key steps of fitting a probability distribution are:

1. Determine a set of potential distributions,
2. Estimate the parameters for the potential distributions, and
3. Assess the distributional fits.

The following sections describe each of these steps in greater detail.

4.4.1 Identification of Probability Distributions

There are several factors to consider when identifying a probability distribution, including:

- *Type of data*: As discussed above, different distributional forms should be considered depending on whether the data are discrete or continuous.
- *Amount of available data*: When limited data are available, additional information (such as historical data, manufacturer's specifications, or expert opinion) may be required before choosing an appropriate distribution. If a large amount of data is available, the results from the EDA (histogram, probability plots) can help inform which distribution might be appropriate.
- *Information about the location, spread, and shape of the distribution*: Using EDA and/or expert judgement to determine whether the data are bounded or if certain values are more probable than others can help inform the choice of distribution.
- *Accuracy required for the distribution*: Distributions for variables that greatly affect the analysis should be chosen with great care. In many metrology applications, accuracy in the tails of the distribution is most important.

Continuous and discrete distributions that are commonly used in metrology are given below.

4.4.1.1 Continuous Distributions

Normal Distribution The normal distribution is an unbounded (in its argument x), symmetric, bell-shaped distribution that is regularly used in metrology. The PDF of the normal (also known as Gaussian) distribution is defined as

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right] \qquad -\infty < x < \infty, \tag{4.5}$$

where μ is the mean and σ is the standard deviation. Figure 4.7 provides a visual depiction of a normal distribution with $\mu = 0$ and $\sigma = 1$. A normal distribution with these parameter values is called a standard normal distribution.

Table 4.2 Characteristics of probability distributions commonly used in metrology

Distribution	Bounded data	Knowledge of probability of values	When to use
Normal	No	Yes	Continuous variables such as voltage, temperature
Rectangular	Yes	No	Calibration interval without additional details regarding probability density
Triangular	Yes	Yes	Bounded interval with data more likely near the midpoint
t-Distribution ($n < 30$)	No	Yes	Computing expanded uncertainties

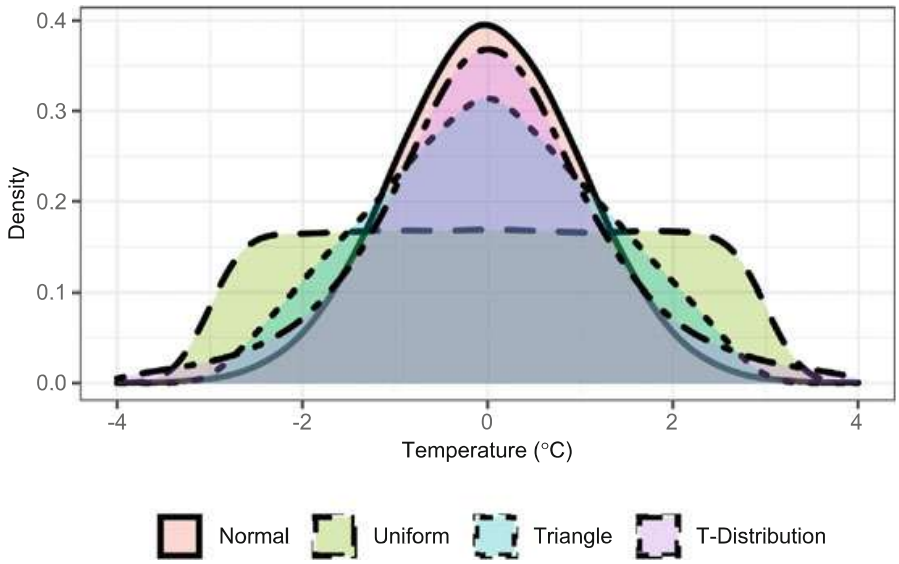


Fig. 4.12 Comparison of normal, uniform, triangular, and t-distributions. Here, the normal distribution has mean 0 and standard deviation 1, the uniform and triangular distributions are bounded on [−3, 3], and the t-distribution has three degrees of freedom

4.4.1.2 Discrete Distributions

Poisson Distribution The Poisson distribution is often used when the data are counts of some event or process occurrence. For example, when counting the number of defects that occur per manufactured part, the Poisson distribution would likely be appropriate (Montgomery 2013). The count of neutrons produced by a laboratory neutron source, analyzed in Sect. 6.4.1, was also modelled using a Poisson distribution.

The Poisson distribution is defined as

$$p(x) = \frac{e^{-\lambda}\lambda^{x}}{x!}, \quad with\ x = 0, 1, 2, 3, \ldots \tag{4.10}$$

The parameter λ, known as the event rate, is both the mean and variance of the distribution. Figure 4.13 provides an example of the Poisson distribution with λ equal to 3. As the value of λ increases, the shape of the distribution becomes more symmetric, and can be approximated by a normal distribution in uncertainty analyses.

4.4.2 Estimating Distribution Parameters

EDA and expert judgement should be used to identify potential distributions. After identification of reasonable distributions for the data, parameters for those distributions are estimated. A commonly used method for estimating parameters of a distribution with moderate to large sample sizes is Maximum Likelihood Estimation (MLE). MLE chooses parameters that are most probable (likely) for a distribution based on the observed data. Many statistical software packages perform MLE. Note that there are many other ways to estimate parameters (particularly if the samples are small), such as the method of moments and Bayesian inference (See Sect. 11.6). For more information on parameter estimation, see Casella and Berger (2002).

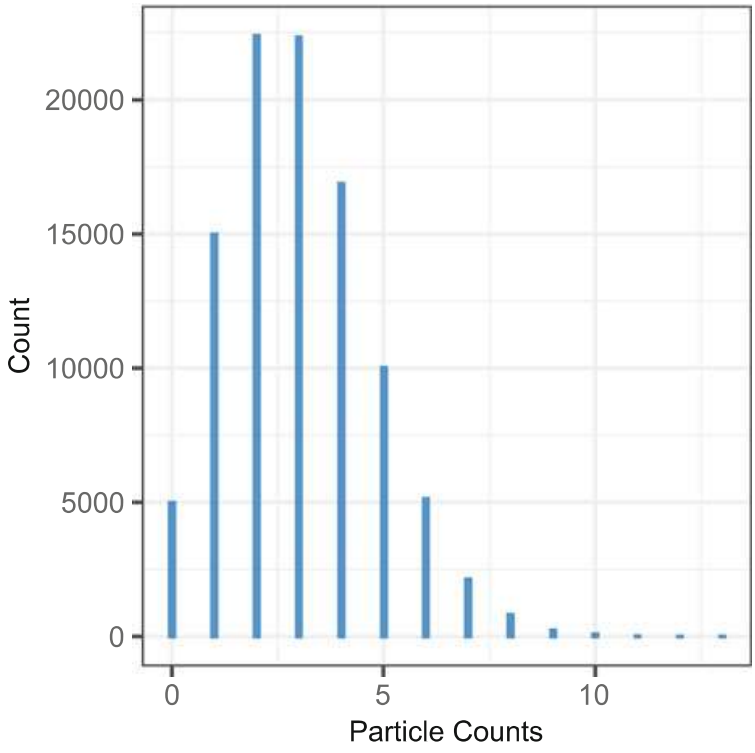


Fig. 4.13 Poisson distribution with mean and variance equal to 3

4.4.3 Assessing Distributional Fit

Goodness of fit diagnostics can be used to compare empirical distributions to different hypothesized distributions after parameters have been estimated. A common diagnostic tool is a probability plot. A probability plot is a visual tool used to assess whether data come from a particular probability distribution. A common probability plot is known as the QQ plot, where the empirical quantiles are plotted against the theoretical quantiles of the hypothesized probability distribution. If the data come from the hypothesized distribution, the points should be approximately linear. If the probability plot is markedly nonlinear, it is evidence that the hypothesized probability distribution is not a good fit.

Figure 4.14 shows how to use the QQ plot as a diagnostic tool. The left and right plots show simulated data generated from normal and Weibull distributions, respectively. The quantiles of the data are compared to the quantiles of a normal distribution. Confidence bounds are also given to show the maximum deviation from the line that should encompass most of the data. In this case, the normal-generated data falls within the confidence bounds, providing evidence that the data are normally distributed. Alternatively, the simulated Weibull data deviate from the fitted line and several points fall outside the confidence bounds, evidence that a normal distribution may not be appropriate.

In metrology applications, assessment of the distributional fit will often be most important in the tails of the distribution. For small sample sizes, it is often difficult to determine which distribution best fits the data, particularly in the tails. Caution should be exercised before choosing a distribution with limited data. Expert judgement or additional data may be needed to properly identify a distribution.

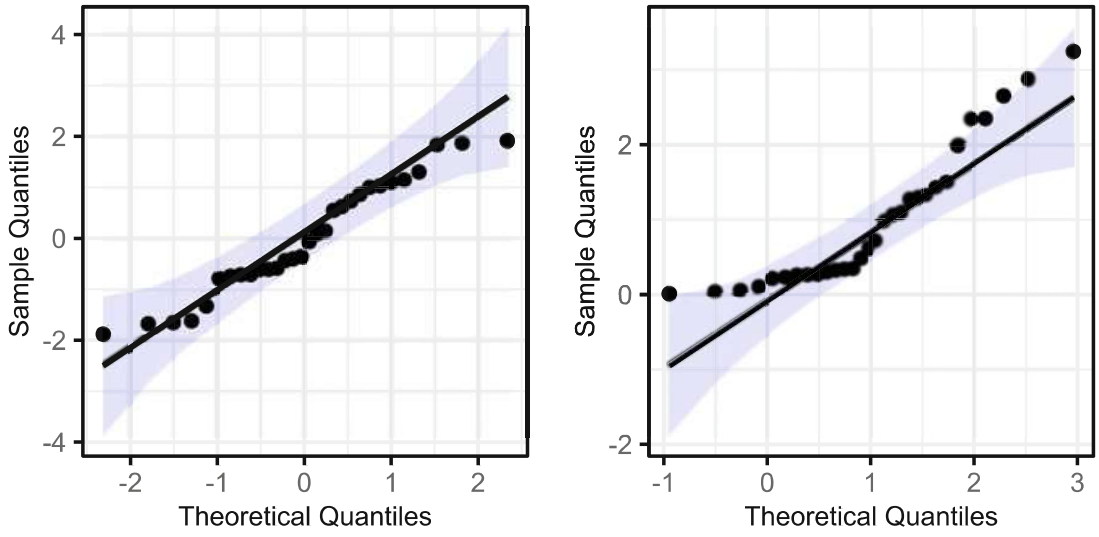


Fig. 4.14 Examples of normal QQ plots used to assess whether data are normally distributed. The left and right plots show data that were generated from normal and Weibull distributions, respectively

When the data are clearly non-normal, the standard uncertainty analysis (Chap. 6) will need to be modified. Monte Carlo (simulation) methods (Chap. 8) may be more appropriate in this case.

4.5 Related Reading

Annex C of the JCGM 100 (2008) (GUM) provides definitions for basic statistical concepts and terms. It gives formal mathematical definitions for some of these concepts along with brief comments on how they might be used in metrology applications. The definitions in the GUM are taken from ISO 3534-1 1993. This ISO provides additional probability and general statistical definitions that are not included in the GUM, including a list of symbols and abbreviations.

The JCGM 101 (2008) (GUM-S1) expands on probability density functions (PDFs) in metrology, including a brief description of how Bayesian methods can be used to estimate a PDF (see Sect. 11.6). The GUM-S1 discusses the principle of maximum entropy for choosing a PDF among a candidate of possible PDFs. Table 1 on p. 20 provides high level guidance on choosing the appropriate PDF based on the type of available information. It includes less-common PDFs such as the curvilinear trapezoid, trapezoidal, and arc sine distributions. It also includes mention of the multivariate Gaussian distribution. The multivariate t-distribution is discussed in the JCGM 102 (2011) (GUM-S2), along with a brief mention of methods for the construction of multivariate PDFs.

Heumann et al. (2016) introduces statistics at an undergraduate level. It provides additional discussion on types of data, methods for analyzing variables using graphs, and summary statistics. It also includes methods for multivariate data including joint and marginal distributions and graphical representations of bi-variate data. Later it discusses elements of probability theory, random variables, and PDFs, with applications of these methods in R.

Additional publications provide an overview of statistical methods in metrology. Chapter 2 of Vardeman and Jobe (2016) begins with basic concepts of statistics along with an introduction to probability modeling. The authors then discuss how one-sample and two-sample statistical methods can be applied to measurement data. The remainder of the chapter briefly discusses several of the concepts covered later in this book, including linear regression (Chap. 9) and Gauge R&R studies (Chap. 10). The National Bureau of Standard's Special Publication 747 (Ku 1988) offers an overview of statistical concepts in metrology along with a postscript on statistical graphics used to check models and their assumptions (e.g. residual plots).

Willink (2006) provides a discussion of principles for probability and statistics in metrology that aid in good decision making. These principles are given with the intent of helping practitioners of metrology produce a statistically defensible analysis. This paper provides high-level considerations for both uncertainty analysis and interlaboratory comparisons.

4.6 Exercises

The data in Table 4.3 provide 15 measurements of temperature, pressure, and number of defects. It will be used for Exercises 4.1–4.8 below.

4.1 Categorize the temperature, pressure, and defects data as continuous or discrete. Further categorize any discrete variables as binary, count or categorical.

4.2 Calculate the following summary statistics for the sample of temperature and pressure data—mean, median, mode, variance, standard deviation.

4.3 What is the standard error of the mean of the pressure data? How can the standard error be interpreted?

4.4 What graphical display(s) are most appropriate for continuous data? For categorical data?

4.5 Generate a histogram of the temperature data. What characteristics do you notice about the spread of these data?

4.6 Generate a boxplot for the pressure data. Are there any outliers present according to the boxplot? If so, what should be done about these outliers?

4.7 Plot the defects data. What distribution would be most appropriate for this type of data?

4.8 Generate a scatterplot for pressure and temperature where pressure is the dependent variable. What do you notice? Now calculate the correlation coefficient for pressure and temperature. How can the correlation coefficient be interpreted?

4.9 Provide an example of when a uniform distribution might be more appropriate than a normal distribution.

4.10 Choose the appropriate probability distribution(s) for the following cases.

a. A manufacturer's specification states that the radius of a cylinder is expected to fall between $\pm 10\%$ of the nominal value.

Table 4.3 Data for Exercises 4.1–4.8

Measurement	Temperature (°F)	Pressure (mmHg)	Defects
1	64.69	645.22	2
2	73.67	728.04	3
3	71.61	717.41	1
4	68.26	678.02	5
5	76.42	761.36	4
6	64.93	644.97	0
7	67.45	678.63	2
8	69.16	687.6	2
9	68.49	689.78	4
10	80.13	814.8	1
11	74.5	744.32	0
12	70.68	706.27	0
13	68.99	683.18	1
14	65.97	661.92	3
15	73.22	737.01	2

Table 4.4 Data for Exercise 4.11

Measurement	Length (m)	Measurement	Length (m)	Measurement	Length (m)
1	1.118	13	0.861	25	1.036
2	1.262	14	1.055	26	1.618
3	1.118	15	0.834	27	1.187
4	0.480	16	1.106	28	0.884
5	1.014	17	0.912	29	0.858
6	0.955	18	0.773	30	1.269
7	0.978	19	0.958	31	0.811
8	0.972	20	1.152	32	0.811
9	1.190	21	1.324	33	0.636
10	1.055	22	2.192	34	0.710
11	0.886	23	0.779	35	0.675
12	1.071	24	0.692	36	0.830

b. The angle of a weld is known to be between 30° and 60°, though angles near 45° are most likely.

c. 50 measurements are taken of the internal pressure of a tank. A plot of the data shows that these measurements are relatively symmetric around the mean value of 50 MPa.

4.11 The data in Table 4.4 show 36 length measurements of a steel plate. Use these data and the statistical software of your choice to complete the following:

a. Calculate the sample mean and standard deviation.

b. Plot a histogram of the data.

c. Use the mean and standard deviation calculated in (a) to fit a normal distribution to the data. Overlay a curve of the normal distribution on the plotted histogram in (b).

d. Generate a QQ plot that compares the data in Table 4.4 to the quantiles of the normal distribution. Does the normal distribution seem appropriate in this case? Why or why not?

References

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JCGM 101 Evaluation of measurement data - Supplement 1 to the guide to the expression of uncertainty in measurement - Propagation of distributions using a Monte Carlo method (2008)

JCGM 102 Evaluation of measurement data - Supplement 2 to the -guide to the expression of uncertainty in measurement - extension to any number of output quantities (2011)